ZJU-UIUC INSTITUTE
Zhejiang University-University of Illinois Urbana-Champaign Institute
浙江大学伊利诺伊大学
厄巴纳香槟校区联合学院

Curriculum
of
Electrical Engineering
For
Class of 2029

Zhejiang University-University of Illinois Urbana-Champaign Institute

(ZJU-UIUC INSTITUTE)

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1. Overview

The Zhejiang University/University of Illinois Urbana-Champaign Institute (ZJU-UIUC Institute, ZJUI) offers a joint dual-degree engineering program. Upon successful completion of the program, and after meeting the graduation requirements of both universities, students will obtain bachelor's degrees separately from Zhejiang University (ZJU) and the University of Illinois Urbana-Champaign (UIUC).

Electrical engineering is a multifaceted discipline that over the last century has produced an astounding progression of technological innovations that have shaped virtually every aspect of modern life. Electrical engineers need a broad and solid foundation in mathematics and physics to support their education in the engineering principles of analysis, synthesis, design, implementation, and testing of the devices and systems that provide the bedrock of modern energy, communication, sensing, computing, medical, security, and defense infrastructures. Within each sub-discipline, there are application domains that strongly rely on hands-on experimental work or that are based on theoretical, mathematical, and computational approaches. The multidisciplinary nature of electrical engineering education addresses the growing demand for the innovation and design of sensing, communication, computing, and decision-making systems of increasing complexity in consumer, defense, and medical applications.

The core curriculum focuses on fundamental courses on circuits, electromagnetics, solid-state electronics, and computer systems, leading to a comprehensive array of specialized courses and laboratories in all of the important areas of modern electrical engineering. These range from power and energy systems to electronic, optoelectronic, and photonic devices; integrated circuits; telecommunications and remote sensing; control systems; robotics; signal processing; and biomedical instrumentation and sensing.

Students are encouraged to take courses and participate in research projects in interdisciplinary areas with Civil and Environmental Engineering, Mechanical Engineering, Computer Engineering, and others during their junior and senior years.

2. Graduation Requirement

2.1 Graduation Requirements for UIUC degree

2.1.1 Grade Point Average Requirement

A student must maintain a minimum GPA of 2.0 (A=4.0) to remain in good standing and graduate. UIUC degree is subjected to the final review by UIUC.

2.1.2 Junior Eligibility Requirement

To be eligible to enroll in the ECE courses listed in the third year of the curriculum, a student must have completed, with a combined 2.25 grade point average, the mathematics, physics, computer science, and electrical and computer engineering courses listed in the first two years.

2.1.3 Curriculum Requirement

The curriculum leading to the degree of Bachelor of Science in Electrical Engineering at UIUC requires **128** academic credits (or academic units, AUs) and is organized into required courses, technical electives, general education, and other electives.

i). Required courses, see section 3 for details.	**Academic Credits**
a) Orientation and Professional Development	1
b) Foundational Mathematics and Science	33
c) Technical Core	28
d) Other Mathematics	3
e) Composition	8
f) Advanced Composition*	4
Total Required AUs:	**77**

ii). Elective courses, see section 4 for details.	
a) Technical Electives	31
b) General Education	12
c) Free Electives	12
Total Required AUs:	**55**

**Students take ECE 445 to satisfy advanced composition and technical core requirement, and credits can be given for both.*

2.1.4 Additional requirements

For UIUC degree, in addition to the specific course and grade point average requirements listed above, each candidate for a bachelor's degree from UIUC must meet the following requirements:

i). Residency Requirement

Earn a minimum sixty (**60**) semester AUs of UIUC credit, of which at least twenty-one (**21**) AUs must be 300 or 400 level UIUC credit courses.

ii). Transfer Requirement

Students should have a satisfactory English Proficiency Test score on TOEFL, IELTS or other exams approved by UIUC, and maintain a good standing on academic studies that all term GPAs, overall GPA on UIUC courses are suggested to be above 2.5. In junior year, students will apply for international transfer to be admitted to UIUC and change academic status from non-degree student to degree student in Fall semester of senior year.

2.2 Graduation Requirements for ZJU degree

In addition to the 128-AU requirement listed above, the curriculum leading to the degree of Bachelor of Engineering from ZJU requires students to complete additional **33.5** academic credits of ZJU-required general education courses for domestic students and additional seven courses (**23** academic credits) in Chinese language and society study for international students. Please refer to section 5 for details.

3. Required Courses

3.1 Orientation and Professional Development

This course introduces students to the opportunities and resources our institute and curriculum can offer you as you work to achieve your career goals. It also provides the skills to work effectively and successfully in the engineering profession.

Course Code	Course Name	Academic Credit
ENGR 100	Engineering Orientation	1
	Total Required AUs:	**1**

3.2 Foundational Mathematics and Science

These courses emphasize the fundamental mathematical and scientific principles upon which the engineering discipline is based.

Course Code	Course Name	Academic Credits
MATH 221	Calculus I	4
MATH 231	Calculus II	3

MATH 241	Calculus III	4
MATH 257	Linear Algebra with Computational Applications	3
MATH 285	Intro to Differential Equations Plus	3
CHEM 102	General Chemistry I	3
CHEM 103	General Chemistry Lab I	1
PHYS 211	University Physics: Mechanics	4
PHYS 212	University Physics: Elec & Mag	4
PHYS 213	University Physics: Thermal Physics	2
PHYS 214	University Physics: Quantum Physics	2
	Total Required AUs:	**33**

3.3 Technical Core

These courses emphasize fundamental concepts and basic laboratory techniques that comprise the common intellectual understanding of electrical engineering.

Course Code	Course Name	Academic Credits
ECE 110	Intro to Electronics	3
ECE 120	Intro to Computing	4
ECE 220	Computer Systems & Programming	4
ECE 210	Analog Signal Processing	4
ECE 329	Fields and Waves Ⅰ	3
ECE 340	Semiconductor Electronics	3
ECE 385	Digital Systems Laboratory	3
ECE 445	Senior Design Project Lab	4
	Total Required AUs:	**28**

3.4 Other Mathematics

This course lays the foundation for understanding problems ranging from communications engineering to data analysis in fields as diverse as medicine and manufacturing. Choose **one (1)** course from the courses in the table.

Course Code	Course Name	Academic Credits

ECE 313	Probability with Engineering Application	3
STAT 410	Statistics and Probability II	3
	Total Required AUs:	3

3.5 Composition

i) These courses teach fundamentals of expository writing.

Course Code	Course Name	Academic Credits
RHET 101	Principles of Writing	4
RHET 102	Principles of Research	4
	Total Required AUs:	**8**

ii) The Advanced Composition requirement is fulfilled by a writing-intensive course beyond basic composition. It is normally taken in the junior or senior year.

Course Code	Course Name	Academic Credits
ECE 445*	Senior Design Project Lab	4
	Total Required AUs:	**4**

**ECE 445 is also a required technical core.*

4. Elective Courses

4.1 Technical Electives

This elective requirement gives each student the freedom to develop a technical course of study in electrical engineering of considerable breadth and focus. The Advanced Core ECE Electives introduce the major sub-disciplines of electrical engineering, such as electrical and computer engineering; bioengineering; acoustics; and magnetic resonance engineering; circuits and signal processing; communication and control; computer engineering; electromagnetics; optics; and remote sensing; microelectronics and quantum electronics; power and energy systems.

Students must complete a minimum of **thirty-one (31)** AUs in technical electives, including at least:

Number of course	Types of elective courses	Course list	Academic Credits
1)	**Non-ECE** courses	section 4.1.1	6

2)	3	**Advanced Core** ECE electives	section 4.1.2	
3)	3	ECE Labs	section 4.1.3	
4)		ECE courses	section 4.1.4	20
			Total required AUs:	**31**

For courses that fall in multiple categories of technical electives, the academic units of the course will be counted to satisfy the AU requirements of each category, but will be counted only once in satisfying the total minimum academic unit requirements (31 AUs) of technical electives.

Technical electives must be taken for a letter grade. S/U graded courses will not count.

4.1.1 Non-ECE Courses

Two (**2**) courses from the following list are required. Students are encouraged to build up their interdisciplinary studies by taking non-ECE technical courses from Civil Engineering, Mechanical Engineering, Computer Science, and other disciplines at ZJUI or during exchange at UIUC.

Aerospace Eng. (AE): 202, 302, 311, 312, 321, 352, 353, 402, 403, 410, 412, 416, 419, 420, 428, 433, 434, 435, 451, 460

Agri. Bio Eng. (ABE): any 300- and 400-level course[1] EXCEPT 440

Astronomy (ASTR): 210, 310, 330, 350, 404, 405, 406, 414, 450

Atmospheric Science (ATMS): 201, 301, 302, 303, 304, 305, 404, 405, 406, 410, 411, 420, 421, 447, 449

Biochemistry (BIOC): 406, 440, 446, 455

Bioengineering (BIOE): 201, 202, 302, 414, 415, 461, 467, 473, 476, 480, 485

Biophysics (BIOP): all 400-level courses[1]

Chem & Bio Eng (CHBE): 221, 321, 421, 422, 424, 430, 431, 440, 451, 452, 453, 456, 457, 471, 472, 473, 474

Chemistry (CHEM): Chem 104, 105, any 200/300/400-level[1] class EXCEPT 396/7, 497, 499

Civil & Env. Eng. (CEE): 310, 330, 408, 410, 416, 430, 447, 491

Crop Sciences (CPSC): 265

Computer Science (CS): 173, 225, 242, 357, 410, 411, 412, 413, 414, 415, 416, 418, 419, 420, 421, 422, 423, 424, 425, 426, 427, 428, 429, 431, 433, 436, 438, 439 ,440, 445, 446, 447, 450, 460, 461, 463, 465, 466, 467, 473, 475, 476; 477, 481, 484.

Geology (GEOL): 107, 208, 333, 380, 411, 417, 432, 436, 440, 450, 452, 460

Industrial Eng. (IE): 310, 330, 360, 361, 400, 410, 411, 412, 413, 420, 430, 431

Integrative Biology (IB): 150, 202, 203, 204, 302, 335, 348, 368, 401, 405, 420, 421, 426, 427, 431, 432, 440, 443, 444, 451, 452, 453, 461, 462, 463, 464, 467, 468, 471, 472, 473, 481, 482, 483, 485, 486

Linguistics (LING): 300, 406, 407, 427

Material Science & Engineering (MSE): 280, any 300/400-level[1] class EXCEPT 304, 460, 461

Math: 213, 347, 348, 357, 402, 403, 412, 413, 414, 415, 416, 417, 418, 423, 424, 425, 427,428, 432, 442, 444, 446, 447, 448, 450, 453, 473, 475, 481, 482, 484, 487, 489, 494

Molecular & Cellular Biology (MCB): 150, 250, 251, 252, 253, 300, 301, 314, 316, 354, 400, 401, 402, 403, 404, 406, 408, 410, 413, 419, 421, 424, 426, 430, 431, 433, 435, 446, 480

Mechanical Eng. (ME): 200, 310, 320, 330, 340, 370, 371, 400, 401, 402, 403, 404, 410, 411, 412, 420, 430, 431, 440, 445, 451, 452, 460, 461, 471, 472, 485, and 487

Music (MUS): 407, 409

Neuroscience (NEUR): 453

Nuclear Plasma & Radiological (NPRE): 201, 247, 402, 412, 421, 423, 429, 431, 432, 435, 441, 442, 444, 446, 447, 448, 451, 455, 457, 458, 470, 475

Physics (PHYS): 225, 246, 325, 326, 370, 371, 401, 402, 403, 406, 419, 420, 427, 460, 466, 470, 485, 486, 487

Psychology (PSYC): 204

Speech & Hearing Science (SHS): 200, 240, 300, 301, 320, 450, and 470

Statistics (STAT): 420, 424, 425, 428, 429, and 440

Systems Eng (SE): 411, 420, 423, 424

Theoretical & Applied Mechanics (TAM): 211, 212, 251, 324, 335, 412, 435, 445, 451

[1] except seminars and special topics courses, which may be reviewed in the Advising Office.

4.1.2 Advanced Core ECE Electives

Select **Three** (**3**) courses from the list below:

Course Code	Course Name	Academic Credits
ECE 310	Digital Signal Processing	3
ECE 330	Power Circuits & Electromechanics	3
ECE 342	Electronic Circuits	3
ECE 350	Fields and Waves II	3

ECE 391	Computer Systems Engineering **or** CS 225 - Data Structure	4

4.1.3 ECE Labs

Students must take **Three (3)** lab courses, at least **one (1)** of which must be a hardware lab.

1) Hardware Labs

Course Code	Course Name	Academic Credits
ECE 343	Electronic Circuits Laboratory	1
ECE 391	Computer Systems Engineering	4
ECE 395	Advanced Digital Projects Lab	2 - 3
ECE 402	Electronic Music Synthesis	3
ECE 415	Biomedical Instrumentation Lab	2
ECE 420	Embedded DSP Laboratory	2
ECE 431	Electric Machinery	4
ECE 435/CS 436	Computer Networking Laboratory	3
ECE 437	Sensors and Instrumentation	3
ECE 438	Communication Networks	3
ECE 439	Wireless Networks	3
ECE 443	LEDs and Solar Cells	4
ECE 444	IC Device Theory	4
ECE 446	Principles of Experimental Research in Electrical Engineering	4
ECE 447	Active Microwave Circuit Design	3
ECE 451	Adv Microwave Measurements	3
ECE 453	Wireless Communication Systems	4
ECE 456	Global Nav Satellite Systems	4
ECE 460	Optical Imaging	3
ECE 463	Digital Communications Lab	2
ECE 466	Optical Communications Lab	1
ECE 468	Optical Remote Sensing	3
ECE 469	Power Electronics Laboratory	2

ECE 470	Introduction to Robotics	4
ECE 481	Nanotechnology	4
ECE 486	Control Systems	4
ECE 489	Robot Dynamics and Control	4
ECE 495	Photonic Device Laboratory	3

2) **Software Labs**

Course Code	Course Name	Academic Credits
ECE 311	Digital Signal Processing Lab	1
ECE 314	Probability in Engineering Lab	1
ECE 364	Programing methods for machine learning	3
ECE 365	Data Science and Engineering	3
ECE 411	Computer Organization & Design	4
ECE 484	Principles of Safe Autonomy	4

4.1.4 ECE courses

Students must take at least **twenty (20)** AUs ECE courses from the list below:

ECE: 304, 307, 310, 311, 314, 330, 333, 342, 343, 350, 364, 365, 374, 380, 391, 395, 402, 403, 407, 408, 410, 411, 412, 414, 415, 416, 417, 418, 419, 420, 421, 422, 424, 425, 428, 431, 432, 434, 435, 437, 438, 439, 441, 443, 444, 446, 447, 448, 449, 451, 452, 453, 454, 455, 456, 457, 458, 459, 461, 460, 462, 463, 464, 465, 466, 467, 468, 469, 470, 471, 472, 473, 476, 478, 479, 480, 481, 482, 483, 484, 485, 486, 487, 488, 489, 490, 491, 492, 493, 495, and 498XX[1].

Course Code	Course Name	Academic Credits
ECE 304	Photonic Devices	3
ECE 307	Techniques for Engineering Decisions	3
ECE 310	Digital Signal Processing	3
ECE 311	Digital Signal Processing Lab	1
ECE 314	Probability in Engineering Lab	1
ECE 330	Power Circuits & Electromechanics	3
ECE 333	Green Electric Energy	3
ECE 342	Electronic Circuits	3
ECE 343	Electronic Circuits Laboratory	1

ECE 350	Fields and Waves II	3
ECE 364	Programming methods for machine learning	3
ECE 365	Data Science and Engineering	3
ECE 374	Introduction to Algorithms & Models of Computation	4
ECE 380	Biomedical Imaging	3
ECE 391	Computer Systems Engineering	4
ECE 395	Advanced Digital Projects Lab	2 or 3
ECE 402	Electronic Music Synthesis	3
ECE 403	Audio Engineering	3
ECE 408	Applied Parallel Programming	4
ECE 411	Computer Organization & Design	4
ECE 412	Microcomputer Laboratory	3
ECE 414	Biomedical Instrumentation	3
ECE 415	Biomedical Instrumentation Lab	2
ECE 416	Biosensors	3
ECE 417	Multimedia Signal Processing	4
ECE 418	Image & Video Processing	4
ECE 419	Security Laboratory	3 or 4
ECE 420	Embedded DSP Laboratory	2
ECE 422	Computer Security I	4
ECE 424	Computer Security II	3 or 4
ECE 425	Intro to VLSI System Design	3
ECE 428	Distributed Systems	3 or 4
ECE 431	Electric Machinery	4
ECE 432	Advanced Electric Machinery	3
ECE 435	Computer Networking Laboratory	3 or 4
ECE 437	Sensors and Instrumentation	3
ECE 438	Communication Networks	3 or 4
ECE 439	Wireless Networks	3 or 4
ECE 441	Physics & Modeling Semiconductor Device	3
ECE 443	LEDs and Solar Cells	4
ECE 444	IC Device Theory & Fabrication	4

ECE 446	Principles of Experimental Research in Electrical Engineering	4
ECE 447	Active Microwave Circuit Design	3
ECE 448	Artificial Intelligence	3 or 4
ECE 449	Machine Learning	3
ECE 451	Adv Microwave Measurements	3
ECE 452	Electromagnetic Fields	3
ECE 453	Wireless Communication Systems	4
ECE 454	Antennas	3
ECE 455	Optical Electronics	3 or 4
ECE 456	Global Nav Satellite Systems	4
ECE 457	Microwave Devices & Circuits	3
ECE 458	Application of Radio Wave Propagation	3
ECE 459	Communications Systems	3
ECE 460	Optical Imaging	4
ECE 461	Digital Communications	3
ECE 462	Logic Synthesis	3
ECE 463	Digital Communications Lab	2
ECE 464	Power Electronics	3
ECE 465	Optical Communications Systems	3
ECE 466	Optical Communications Lab	1
ECE 467	Biophotonics	3
ECE 468	Optical Remote Sensing	3
ECE 469	Power Electronics Laboratory	2
ECE 470	Introduction to Robotics	4
ECE 472	Biomedical Ultrasound Imaging	3
ECE 473	Fund of Engineering Acoustics	3 or 4
ECE 476	Power System Analysis	3
ECE 478	Formal Software Development Methods	3 or 4
ECE 479	IoT and Cognitive Computing	4
ECE 480	Magnetic Resonance Imaging	3 or 4
ECE 481	Nanotechnology	4

ECE 482	Digital IC Design	3
ECE 483	Analog IC Design	3
ECE 485	MEMS Devices & Systems	3
ECE 486	Control Systems	4
ECE 487	Introduction Quantum Electronics for EEs	3
ECE 488	Compound Semiconductor & Devices	3
ECE 489	Robot Dynamics and Control	4
ECE 490	Introduction to Optimization	3 or 4
ECE 491	Numerical Analysis	3 or 4
ECE 492	Parallel Program: Science & Engineering	3 or 4
ECE 493	Advanced Engineering Math	3 or 4
ECE 495	Photonic Device Laboratory	3

[1] ECE 498XX refers to a series of ECE courses developed by ZJUI. Consult UA office at ZJUI for a complete course list.

4.2 General Education

The general education courses develop students' understanding of human culture and society, build skills of inquiry and critical thinking, and lay a foundation for civic engagement and lifelong learning. To satisfy the General Education requirements, students must complete a minimum of **twelve (12)** academic credits in general education courses. Students must take all courses for grade and complete sufficient courses in required categories based on the table below:

Required course category	Number of courses	Academic Credits
1) **Humanities & Arts**	Two (2)	6
2) **Social & Behavioral Sciences**[1]	Two (2)	6
Additional Requirements:		
i) Culture Studies[2]		
Western/Comparative Culture(s)	One (1)	3
Non-Western Culture(s)	One (1)	3
U.S. Minority Culture(s)[3]	One (1)	3
ii) Aesthetic Education[4]	One (1)	3
	Total required AUs:	**12**

[1] One of the Social and Behavioral Science courses is recommended to be an introductory economics course (ECON 102 or ECON 103).

[2] Proper selection of Social and Behavioral Sciences and in Humanities and the Arts courses will assure that these courses also fulfill Western, non-Western and US Minority Cultures requirements.

[3] Not required if students pursue ZJU degree only.

[4] Required only for ZJU degree.

For courses that fall in multiple categories of general education, the academic units of the course will be counted to satisfy the AU requirements of each category, but be counted only once in satisfying the total minimum academic unit requirements (12 AUs) of general education.

4.3 Free Electives

These unrestricted electives give students the opportunity to explore any intellectual area of unique interest. This freedom plays a critical role in helping students define their research specialties. Students are encouraged to take cross-disciplinary courses as free electives.

Students must complete a minimum of **twelve (12)** AUs in free electives.
ZJU required general education courses cannot be counted as free electives to satisfy with the requirements of UIUC degree.

5. ZJU Required General Education

These courses introduce modern Chinese history, social development, government policies, etc., and help students improve their English and maintain a healthy lifestyle.

5.1 For domestic students

Domestic students must complete all the courses below to fulfill the graduation requirements for the Bachelor of Engineering Degree at ZJU. These courses can be taught in Chinese. The academic credits in this section are additional to 128 AUs of credits required for UIUC degree.

Course Code	Course Name	Academic Credits
LAW 1002	Ideology, Morality and Rule of Law	3
HIST 2001	Modern Chinese History	3
PHIL 2002	Introduction to the Principle of Marxism	3
PS 2011	Intro.to Mao Thought & Theoretical System of China Socialism	3

Course Code	Course Name	Academic Credits
PS 3011	General Introduction to Xi Jinping Thought on Socialism with Chinese Characteristics for a New Era	3
PS 1001	Situation and Policy I	1
PS 2001	Situation and Policy II	1
ENGL 1001	Integrated English I	3
ENGL 1002	Integrated English II	3
PE 1001	Physical Education I	1
PE 1002	Physical Education II	1
PE 2001	Physical Education III	1
PE 2002	Physical Education IV	1
PE 4021	Physical Education VII-Fitness test and exercise	0.5
MITR 1001	Military Training	2
MITR 2001	Military Theory	2
ST 4001	Senior Thesis	2
	Total Required AUs:	**33.5**

5.2 For International students

International students are required to complete the following courses in Chinese language study and Chinese society to fulfill the graduation requirements along with the above 128 academic credits of courses towards the Bachelor of Engineering Degree at ZJU.

Course Code	Course Name	Academic Credits
CHIN 1001	Chinese I	4
CHIN 1002	Chinese II	4
CHIN 1003	Chinese III	4
CHIN 1004	Chinese IV	4
CHIN 1005	International Chinese Language Proficiency and Testing	2
CULT 2001	China Survey	3
ST 4001	Senior Thesis	2
	Total Required AUs:	**23**

6. Course Schedule Sample by Semester

6.1 First Year-First (Fall) Semester

List of ZJUI/UIUC courses for both domestic and international students

No	Course Code	Course Name	Academic Credits
1	Rhet 101	Principles of Writing	4
2	Chem 102	General Chemistry I	3
3	Chem 103	General Chemistry Lab I	1
4	Math 221	Calculus I	4
5	ECE 110	Intro to Electronics	3
6	Engr 100	Engineering Orientation	1
		Total:	16

** Students would take MATH 231 if MATH 221 is exempted.*
*** Students can select one more course, such as CS 101 or MATH 213 if MATH 221 is exempted.*

List of ZJU courses for domestic students

No	Course Code	Course Name	Academic Credits
1	MITR 1001	Military Training	2
2	ENGL 1001	Integrated English I	3
3	PHIL 2002	Introduction to the Principle of Marxism	3
4	PS 2011	Intro.to Mao Thought & Theoretical System of China Socialism	3
5	PE 1001	Physical Education I	1
6	PS 1001	Situation and Policy I	

List of ZJU courses for international students

No	Course Code	Course Name	Academic Credits
1	CHIN1001	Chinese I	4

6.2 First Year-Second (Spring) Semester

List of ZJUI/UIUC courses for both domestic and international students

No	Course Code	Course Name	Academic Credits
1	Rhet 102	Principles of Research	4
2	Math 231	Calculus II	3

3	Phys 211	University Physics: Mechanics	4
4	ECE 120	Intro to Computing	4
5	MATH 257	Linear Algebra with Computational Application	3
		Total:	18

List of ZJU Courses for domestic students

No	Course Code	Course Name	Academic Credits
1	LAW1002	Ideology, Morality and Rule of Law	3
2	ENGL1002	Integrated English II	3
3	PE1002	Physical Education II	1
4	PS 1001B	Situation and Policy I	1

List of ZJU Courses for international students

No	Course Code	Course Name	Academic Credits
1	CHIN1002	Chinese II	4

6.3 Second Year-First (Fall) Semester

List of ZJUI/UIUC courses for both domestic and international students

No	Course Code	Course Name	Academic Credits
1	Math 241	Calculus III	4
2	Phys 212	University Physics: Elec& Mag	4
3	ECE 220	Computer Systems & Programming	4
4	Math 213[1]	Basic Discrete Mathematics	3
5	GenEd 1[2]	General Education Elective	3
		Total:	18

[1] Technical electives for EE students;

[2] Suggest selecting ECON 102 or ECON 103.

List of ZJU Courses for domestic students

No	Course Code	Course Name	Academic Credits
1	PS 3011	General Introduction to Xi Jinping Thought on Socialism with Chinese Characteristics for a New Era	3
2	PE 2001	Physical Education III	1
3	MITR 2001	Military Theory	2

4	PS 2001	Situation and Policy II

List of ZJU Courses for international students

No	Course Code	Course Name	Academic Credits
1	CHIN1003	Chinese III	4

6.4 Second Year-Second (Spring) Semester

List of ZJUI/UIUC courses for both domestic and international students

No	Course Code	Course Name	Academic Credits
1	Math 285	Intro to Differential Eq	3
2	Phys 213	Univ Physics: Thermal Physics	2
3	Phys 214	Univ Physics: Quantum Physics	2
4	ECE 210	Analog Signal Processing	4
5	CS 225*	Data Structure	4
6	GenEd 2	General Education Elective	3
		Total:	18

* Technical electives for EE students.

List of ZJU Courses for domestic students

No	Course Code	Course Name	Academic Credits
1	HIST 2001	Modern Chinese History	3
2	PS 2001	Situation and Policy II	
3	PE 2002	Physical Education IV	1

List of ZJU Courses for international students

No	Course Code	Course Name	Academic Credits
1	CHIN1004	Chinese IV	4
2	CHIN 1005	International Chinese Language Proficiency and Testing	2

6.5 Third Year-First (Fall) Semester

Juniors exchange to UIUC (dual degree only)

No	Course Code	Course Name	Academic Credits

1	ECE 329	Fields and Waves I	3
2	ECE 385	Digital Systems Laboratory	3
3	Tech Elec	One of Advanced Core ECE Electives	3
4	Tech Elec	One of Advanced Core ECE Electives	3
5	GenEd 3	General Education Elective	3
		Total:	16

6.6 Third Year- Second (Spring) Semester

No	Course Code	Course Name	Academic Credits
1	ECE 313/314	Probability with Engrg Applic/Lab	3+1
2	ECE 340	Semiconductor Electronics	3
3	Tech Elec	One of Advanced Core ECE Electives	3
4	Tech Elec	Technical Elective	3
5	GenEd 4	General Education Elective	3
		Total:	16

6.7 Fourth Year-First (Fall) Semester

No	Course Code	Course Name	Academic Credits
1	Tech Elec	Technical Elective	3
2	Tech Elec	Technical Elective	3
3	Tech Elec	Technical Elective	3
4	Free Elec	Free Elective	3
5	Free Elec	Free Elective	3
		Total:	15

List of ZJU Courses for domestic students

No	Course Code	Course Name	Academic Credits
1	ST 4001	Senior thesis	
2	PS 2001	Situation and Policy II	

List of ZJU Courses for international students

No	Course Code	Course Name	Academic Credits
1	ST 4001	Senior Thesis	

6.8 Fourth Year- Second (Spring) Semester

List of ZJUI/UIUC courses for both domestic and international students

No	Course Code	Course Name	Academic Credits
1	ECE 445	Senior Design Project Lab	4
2	Tech Elec	Technical Elective	3
3	Tech Elec	Technical Elective	3
4	Free Elec	Free Elective	3
		Total:	13

List of ZJU Courses for domestic students

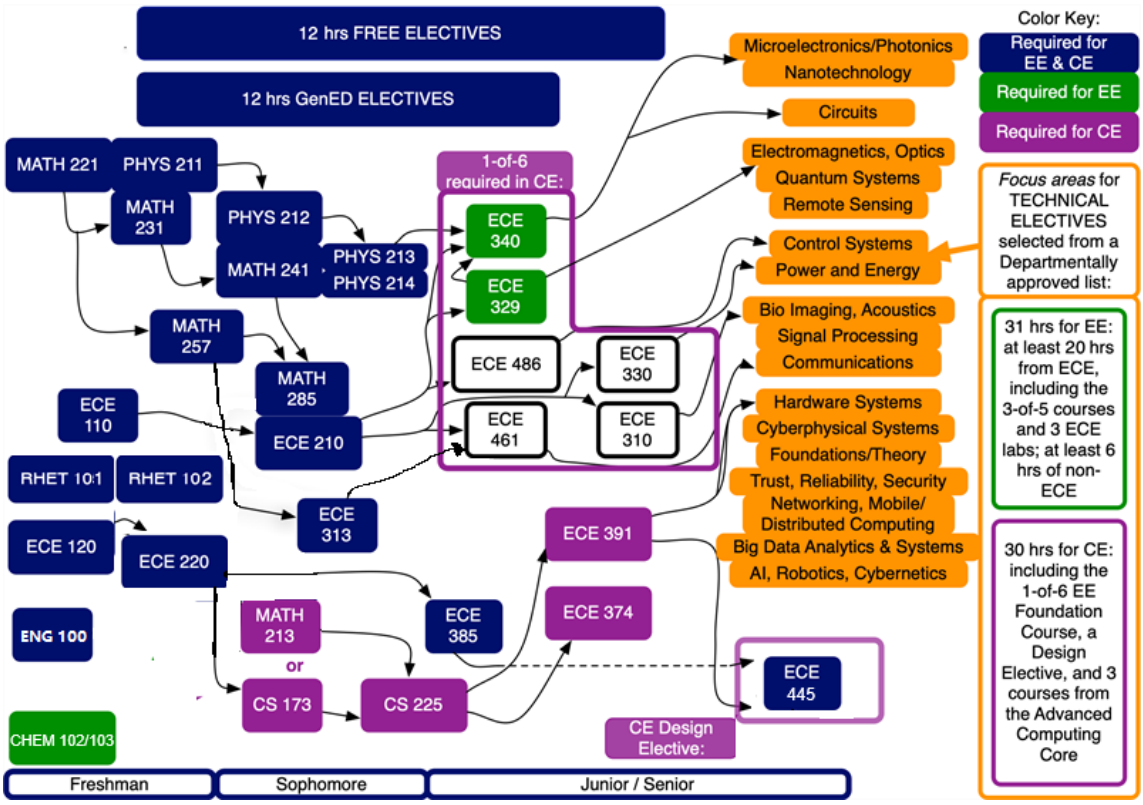
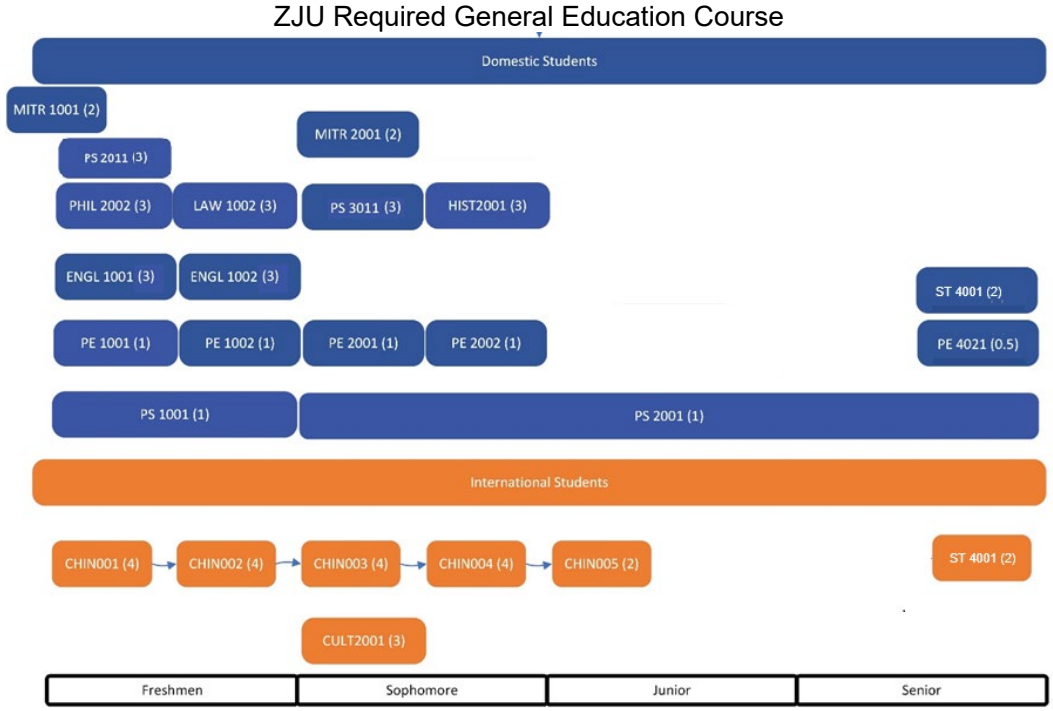
No	Course Code	Course Name	Academic Credits
1	PE 4021	Physical Education VII-Fitness test and exercise	0.5
2	PS 2001F	Situation and Policy II	1
3	ST 4001	Senior Thesis	2

List of ZJU Courses for international students

No	Course Code	Course Name	Academic Credits
1	ST 4001	Senior Thesis	2

7. Curriculum Flow Map

The following flow map provides a quick summary of the main features of the Electrical Engineering curriculum.



ZJU Required General Education Course